Ex – 06	EDA – Data Visualization with Matplotlib

AIM

To perform Exploratory Data Analysis (EDA) using Matplotlib by creating line charts, bar charts, and histograms for the Student Performance dataset.

PROBLEM STATEMENT

To visualize student performance data using line charts, bar charts, and histograms to understand score trends, compare average subject marks, and analyze mark distributions.

PROCEDURE

1. Import the Pandas and Matplotlib libraries.
2. Load the Student Performance dataset using Pandas.
3. Select the Mathematics, Reading, and Writing score columns.
4. Check and handle missing values in the selected columns.
5. Create a line chart to display student marks across selected records.
6. Add a title, axis labels, and a legend to the line chart.
7. Create a bar chart to compare average marks across subjects.
8. Create a histogram to display the distribution of student marks.
9. Customize the charts with colors, labels, and appropriate figure sizes.
10. Display the charts and analyze the visualization results.

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Step 1: Load the dataset
df = pd.read_csv('Student_Perfomance.csv')
```

```
# Select subject columns
subjects = ['math score', 'reading score', 'writing score']

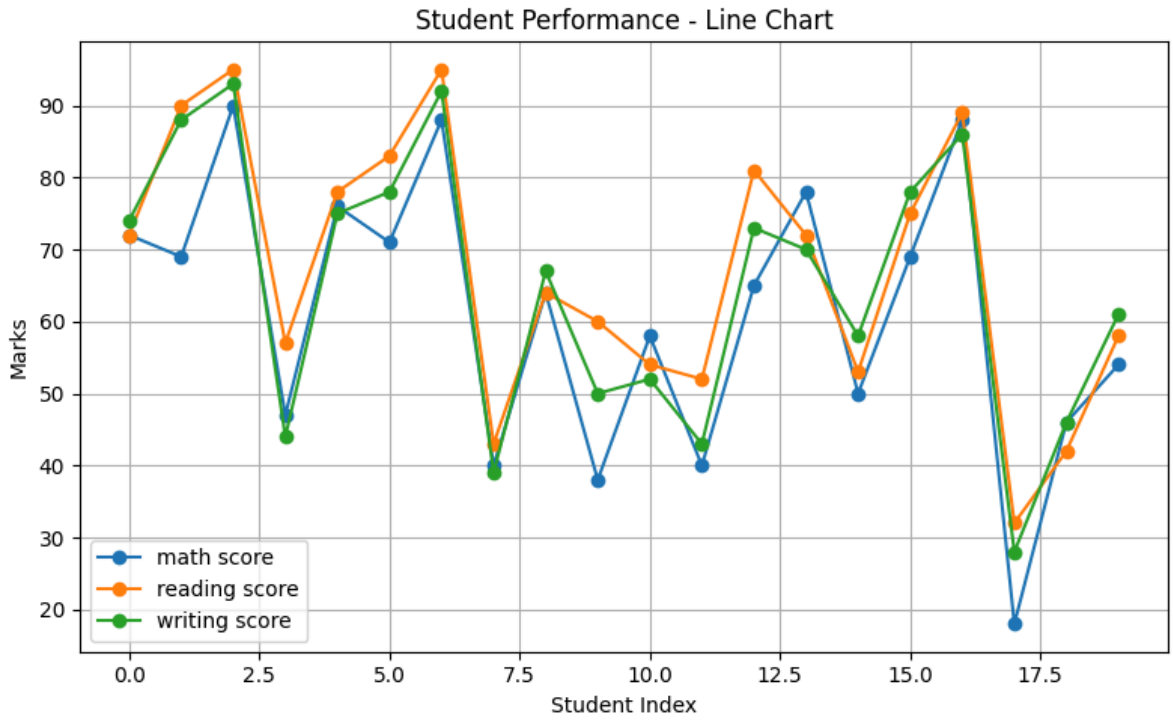
# Convert scores to numeric and handle missing values
df[subjects] = df[subjects].apply(
    pd.to_numeric, errors='coerce'
)
df[subjects] = df[subjects].fillna(df[subjects].mean())
```

Line Chart - Student Marks

```
# Step 2: Line Chart - Student Marks
plt.figure(figsize=(8, 5))

for subject in subjects:
    plt.plot(df.index[:20], df[subject].iloc[:20],
             marker='o', label=subject)

plt.title('Student Performance - Line Chart')
plt.xlabel('Student Index')
plt.ylabel('Marks')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

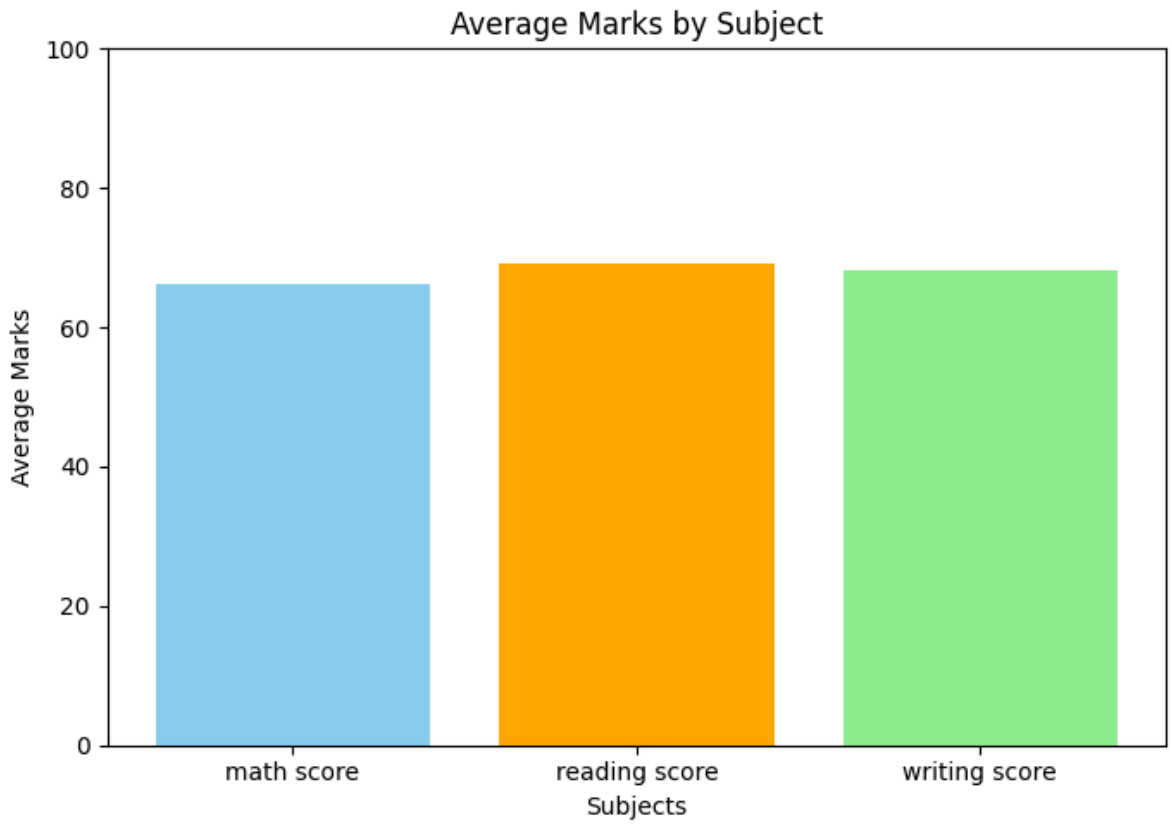


Bar Chart - Average Marks

```
# Step 3: Bar Chart - Average Marks
average_scores = df[subjects].mean()

plt.figure(figsize=(7, 5))
plt.bar(average_scores.index, average_scores.values,
        color=['skyblue', 'orange', 'lightgreen'])

plt.title('Average Marks by Subject')
plt.xlabel('Subjects')
plt.ylabel('Average Marks')
plt.ylim(0, 100)
plt.tight_layout()
plt.show()
```

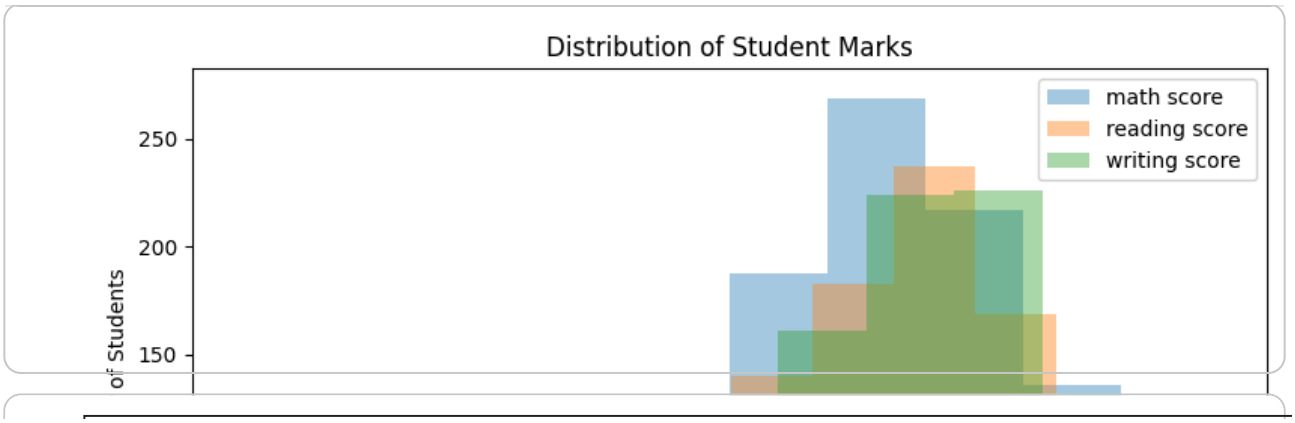


Histogram - Distribution of Marks

```
# Step 4: Histogram - Distribution of Marks
plt.figure(figsize=(8, 5))

for subject in subjects:
    plt.hist(df[subject], bins=10, alpha=0.4,
             label=subject)

plt.title('Distribution of Student Marks')
plt.xlabel('Marks')
plt.ylabel('Number of Students')
plt.legend()
plt.tight_layout()
plt.show()
```


Distribution of Student Marks
math score
reading score
writing score
250
200
150
of Students

OUTPUT

The program generates three visualizations:

1. Line chart: Displays the marks of the first 20 students across the three subjects.
2. Bar chart: Compares the average Mathematics, Reading, and Writing scores.
3. Histogram: Shows the distribution of marks in the three subjects.

RESULT

EDA was performed successfully using Matplotlib. Line charts, bar charts, and histograms were created to visualize student performance trends, compare average scores, and analyze mark distributions.